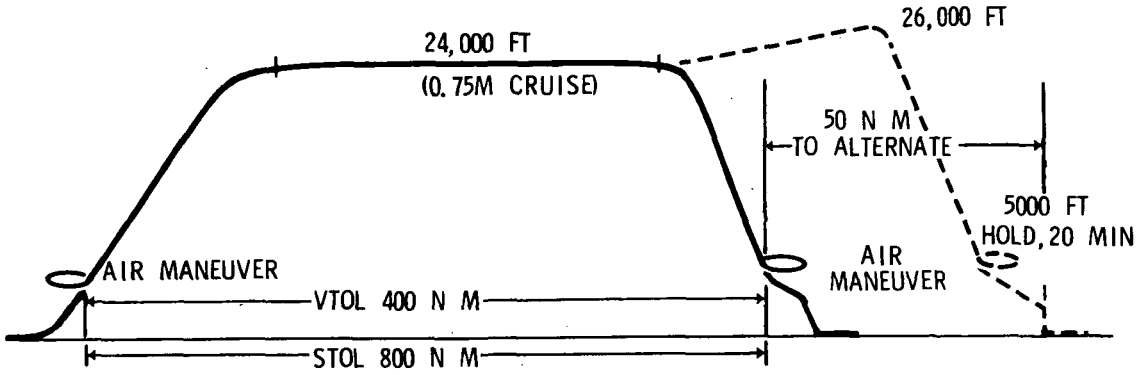
Performance

The selected aircraft configuration was sized to meet the design mission range/payload requirements at a cruise speed of 0.75 Mach number. Figure 19 presents the leg data for the design missions. The cruise altitude is 24,000



OPER MODE	PARAMETER		TAXI & TAKEOFF	AIR MANEUVER	CLIMB	CRUISE	DESCEND	AIR MANEUVER	APPROACH & LAND	RESERVE LEGS	TRIP TOTALS*
VTOL	TIME	MIN	1.5	0.5	15.2	36.6	4.8	1.5	2.9	-	63.0 MIN
	DIST	%	0	0	21.3%	69.0%	9.8	0	0	-	400 N M
	FUEL	%	7.7%	0.4%	19.9%	39.4%	2.1%	1.2%	6.8%	32.5%	14,733 LB
STOL	TIME	MIN	2.5	1.0	21.1	84.6	4.8	3.0	3.9	-	120.9 MIN
	DIST	%	0	0	15.4%	79.9%	4.9%	0	0	-	800 N M
	FUEL	%	5.8%	0.6%	16.7%	58.1%	1.3%	1.5%	5.3%	19.9%	24,075 LB

*TIME & DISTANCE FOR REVENUE TRIP, FUEL FOR THRU FLIGHT TO ALTERNATE

Figure 19. Design Mission Profile Performance

feet for both missions. This altitude is sufficient to avoid the majority of severe turbulence, icing and other undesirable features of the lower regions of the atmosphere. The STOL takeoff distance at the STOL takeoff weight of 110,020 pounds after a critical failure is 1240 feet to the 35 foot obstacle. Data on the VTOL takeoff trajectories are presented in the noise trade study subsection of the report.

Economic Analysis

The economic analysis of the selected airplane consisted of evaluation of the system direct operating cost (DOC) characteristics. The analysis was performed using the methodology of Reference 3 and constant factors provided by the study guidelines for propulsion and airframe costs and selected other parameters. Propulsion system costs are based upon a 20,000 pound thrust unit costing $690,000. Of this price 58% ($400,000) is assumed to be fan cost and 42% ($290,000) assumed to be gas generator cost. The costs are scaled for alternate sizes of propulsion system elements using the data of figure 20.